

Overfitted-Dimension Identifiability for One and Two MiSo Subcones

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Abstract

We first study one MiSo subcone. The marginal count distribution identifies its distinct normalized factor directions, Gamma rates, and total Gamma shape attached to each direction-rate pair. An overfitted representation can differ from the minimal representation only by Gamma splitting or by a null boundary dimension. We then treat two mixture components. Distinct nonnested cones are identified as separate conic strata of the recovered Poisson mixing measure. When one cone is strictly nested inside another with lower dimension, the lower-dimensional component is instead identified as the unique Hausdorff-singular part in the Lebesgue decomposition of that mixing measure. In either case, the one-subcone result applies after the components are separated.

1 Single-subcone Gamma–Poisson model

Let $Y_i = (Y_{i1}, \dots, Y_{iM})^\top$ be independent observations satisfying

$$Y_{im} \mid \mu_i \sim \text{Poi}(\mu_{im}), \quad m = 1, \dots, M, \quad (1)$$

$$\mu_i = \sum_{d=1}^{D_*} \lambda_{id} f_d^*. \quad (2)$$

Conditional on μ_i , the coordinates of Y_i are independent. The factor directions are normalized,

$$f_d^* \in \mathbb{R}_+^M, \quad \mathbf{1}^\top f_d^* = 1,$$

and the loadings are mutually independent with the shape–rate parameterization

$$\lambda_{id} \sim \text{Gamma}(a_d^*, b_d^*), \quad a_d^* > 0, \quad b_d^* > 0.$$

In the MiSo interpretation, f_d^* is the factor direction selected by dimension d . The word *dimension* is used throughout; no separate “slot” terminology is needed.

We compare the true model with an overfitted representation

$$\tilde{\mu}_i = \sum_{h=1}^D \tilde{\lambda}_{ih} \tilde{f}_h, \quad D \geq D_*, \quad (3)$$

where

$$\tilde{\lambda}_{ih} \sim \text{Gamma}(\tilde{a}_h, \tilde{b}_h), \quad \tilde{a}_h, \tilde{b}_h > 0, \quad \tilde{f}_h \in \mathbb{R}_+^M, \quad \mathbf{1}^\top \tilde{f}_h = 1.$$

The fitted loadings are also mutually independent.

Define the scaled directions

$$q_d^* = \frac{f_d^*}{b_d^*}, \quad \tilde{q}_h = \frac{\tilde{f}_h}{\tilde{b}_h}. \quad (4)$$

Normalization makes this reparameterization one-to-one because

$$b = \frac{1}{\mathbf{1}^\top q}, \quad f = \frac{q}{\mathbf{1}^\top q}. \quad (5)$$

Assumption 1 (Minimal true representation). The vectors $q_1^*, \dots, q_{D_*}^*$ are pairwise distinct. When D_* is interpreted as the linear dimension of the subcone, we additionally assume that $f_1^*, \dots, f_{D_*}^*$ are linearly independent.

Pairwise distinctness is the condition needed for statistical identification. Linear independence is stronger and connects the statistical result to the geometric dimension of a simplicial subcone.

Relation to existing identifiability theory

Three established lines of theory meet in this result. First, a mixed Poisson law encodes the Laplace transform of its mixing distribution; the identifiability of Poisson mixtures is classical [9, 3]. Second, Gamma laws form a convolution semigroup at a common rate and are infinitely divisible [8]. Third, ordinary NMF uniqueness is a geometric question and usually requires additional conditions on the factorization [6, 2]. Theorem 1 is not a direct restatement of any one of these results: it uses the full Gamma–Poisson distribution to identify the finite set of scaled directions and then characterizes exactly the remaining overfitting ambiguity. In particular, its conclusion is distributional rather than uniqueness of one realized matrix factorization.

2 Overfitted identifiability

Theorem 1 (Identifiability up to Gamma splitting). *Suppose Assumption 1 holds. If the true model (2) and the overfitted model (3) induce the same distribution for Y_i , then the distinct scaled directions in the fitted model equal those in the true model:*

$$\{\tilde{q}_h : h = 1, \dots, D\}_{\text{distinct}} = \{q_d^* : d = 1, \dots, D_*\}.$$

Moreover, for every $d = 1, \dots, D_*$,

$$\sum_{h: \tilde{q}_h = q_d^*} \tilde{a}_h = a_d^*. \quad (6)$$

Consequently, the fitted dimensions can be partitioned into nonempty groups $G_1, \dots, G_{D_*}$ such that, for every $h \in G_d$,

$$\begin{aligned} \tilde{f}_h &= f_d^*, & \tilde{b}_h &= b_d^*, \\ \sum_{h \in G_d} \tilde{a}_h &= a_d^*. \end{aligned} \quad (7)$$

Thus the minimal direction-rate pairs and their total shapes are identifiable up to permutation, although the separate shapes within an overfitted group are not identifiable.

Proof. For $z \in [0, 1]^M$, the probability-generating function of the count vector is

$$G_Y(z) = \mathbb{E} \left[\prod_{m=1}^M z_m^{Y_{im}} \right] \quad (8)$$

$$= \mathbb{E}[\exp\{-\langle \mathbf{1} - z, \mu_i \rangle\}] = \mathcal{L}_\mu(\mathbf{1} - z), \quad (9)$$

where $\mathcal{L}_\mu(t) = \mathbb{E}\{\exp(-\langle t, \mu_i \rangle)\}$ is the multivariate Laplace transform of the random Poisson mean. Hence equality of the count laws implies equality of the Laplace transforms of μ_i and $\tilde{\mu}_i$ on a nonempty open set. Uniqueness of the multivariate Laplace transform then gives equality of their mixing distributions. This is the transform argument behind classical identifiability results for Poisson mixtures [9]; see also the monograph treatment of mixed Poisson models in [3].

For a shape–rate Gamma variable,

$$\mathbb{E}\{e^{-u\lambda}\} = \left(\frac{b}{b+u}\right)^a.$$

Using independence across dimensions and (4),

$$\mathcal{L}_\mu(t) = \prod_{d=1}^{D_*} (1 + \langle q_d^*, t \rangle)^{-a_d^*}. \quad (10)$$

Define the negative log-Laplace transform

$$\Psi(t) = -\log \mathcal{L}_\mu(t) = \sum_{d=1}^{D_*} a_d^* \log(1 + \langle q_d^*, t \rangle).$$

The fitted representation analogously gives

$$\tilde{\Psi}(t) = \sum_{h=1}^D \tilde{a}_h \log(1 + \langle \tilde{q}_h, t \rangle).$$

Equality of the count laws implies $\Psi = \tilde{\Psi}$ on an open set. Differentiating yields equal vector-valued rational functions,

$$\sum_{d=1}^{D_*} \frac{a_d^* q_d^*}{1 + \langle q_d^*, t \rangle} = \sum_{h=1}^D \frac{\tilde{a}_h \tilde{q}_h}{1 + \langle \tilde{q}_h, t \rangle}. \quad (11)$$

Group identical scaled directions on each side. A distinct vector q creates the singular hyperplane

$$\mathcal{H}_q = \{t : 1 + \langle q, t \rangle = 0\}.$$

Choose a generic point on $\mathcal{H}_q$ that lies on none of the other singular hyperplanes. Multiplying (11) by $1 + \langle q, t \rangle$ and approaching this point leaves the nonzero vector

$$\left(\sum_{d: q_d^* = q} a_d^* \right) q$$

on the true side. Since all shapes are positive, this singularity cannot cancel. The fitted side must therefore contain the same hyperplane. Two hyperplanes of the form $1 + \langle q, t \rangle = 0$ are equal only when their vectors q are equal. Repeating the argument in both directions proves equality of the sets of distinct scaled directions.

Matching the nonzero limits at each common hyperplane gives

$$\left(\sum_{h: \tilde{q}_h = q_d^*} \tilde{a}_h \right) q_d^* = a_d^* q_d^*.$$

Since $q_d^* \neq 0$, this proves (6). Finally, (5) shows that equality of scaled directions implies equality of both the normalized factor directions and the Gamma rates, proving (7). $\square$

Corollary 2 (Identification without duplicate dimensions). *Under the conditions of Theorem 1, if the fitted model is constrained so that $\tilde{q}_1, \dots, \tilde{q}_D$ are pairwise distinct, equality of the count distributions implies*

$$D = D_*,$$

and all parameters agree up to a permutation of the dimensions.

Proof. Theorem 1 gives exactly D_* distinct fitted scaled directions. If every fitted direction is distinct, their number is D , hence $D = D_*$. Equation (6) then matches the corresponding shape parameters. $\square$

3 Two-subcone mixtures

Let $Z_i \in \{1, 2\}$ with

$$\Pr(Z_i = s) = \pi_s, \quad \pi_s > 0, \quad \pi_1 + \pi_2 = 1.$$

Conditional on $Z_i = s$, suppose

$$\mu_i = \sum_{d=1}^{D_s^*} \lambda_{isd} f_{sd}^*, \quad \lambda_{isd} \sim \text{Gamma}(a_{sd}^*, b_{sd}^*) \quad \text{independently,} \quad (12)$$

where $a_{sd}^*, b_{sd}^* > 0$, $f_{sd}^* \in \mathbb{R}_+^M$, and $\mathbf{1}^\top f_{sd}^* = 1$. Define

$$q_{sd}^* = \frac{f_{sd}^*}{b_{sd}^*}, \quad \mathcal{C}_s^* = \text{cone}(f_{s1}^*, \dots, f_{sD_s^*}^*), \quad V_s^* = \text{span}(\mathcal{C}_s^*).$$

Let ν_s^* denote the conditional law of μ_i given $Z_i = s$ and

$$\nu^* = \pi_1^* \nu_1^* + \pi_2^* \nu_2^*$$

its unconditional mixing law. We assume within each true component that the q_{sd}^* are pairwise distinct and the f_{sd}^* are linearly independent. The corresponding count law identifies ν^* : equation (9) recovers its multivariate Laplace transform on an open set, and uniqueness of that transform recovers the probability measure.

Lemma 3 (Gamma measure on a simplicial cone). *For $s \in \{1, 2\}$, the measure ν_s^* has support $\mathcal{C}_s^*$, assigns probability one to $\text{relint}(\mathcal{C}_s^*)$, and has a strictly positive density there with respect to D_s^* -dimensional Hausdorff measure on V_s^* . In particular, every subset of V_s^* having Hausdorff dimension smaller than D_s^* has ν_s^* -measure zero.*

Proof. Write $F_s^* = [f_{s1}^* \cdots f_{sD_s^*}^*]$. Linear independence makes $x \mapsto F_s^* x$ a linear homeomorphism from $(0, \infty)^{D_s^*}$ onto $\text{relint}(\mathcal{C}_s^*)$. The independent Gamma vector has a strictly positive Lebesgue density on $(0, \infty)^{D_s^*}$. Its pushforward therefore has a strictly positive density relative to Hausdorff measure on the image. The closure of that image is $\mathcal{C}_s^*$, which proves the support claim. These standard Hausdorff-measure facts are reviewed in [7]. $\square$

Theorem 4 (Two distinct nonnested subcones). *Suppose*

$$V_1^* \not\subseteq V_2^*, \quad V_2^* \not\subseteq V_1^*. \quad (13)$$

Consider any other two-component MiSo representation having positive mixture weights, proper Gamma loadings, and component laws that are nondegenerate on the relative interiors of their fitted

cones. If it induces the same count law as the true model, then, after possibly swapping the two component labels,

$$\tilde{\pi}_s = \pi_s^*, \quad \tilde{\nu}_s = \nu_s^*, \quad \tilde{\mathcal{C}}_s = \mathcal{C}_s^*, \quad s = 1, 2. \quad (14)$$

Within each matched component, the fitted dimensions satisfy the conclusion of Theorem 1: they can only duplicate a true direction-rate pair and split its Gamma shape, apart from dimensions on a null boundary.

Proof. Equality of the count laws implies equality of the mixing laws. By Lemma 3,

$$\nu_1^*(V_2^*) = 0, \quad \nu_2^*(V_1^*) = 0,$$

because $V_1^* \cap V_2^*$ has dimension strictly smaller than both D_1^* and D_2^* . Thus the two true component laws are mutually singular, with full-measure strata

$$A_1 = \text{relint}(\mathcal{C}_1^*) \setminus V_2^*, \quad A_2 = \text{relint}(\mathcal{C}_2^*) \setminus V_1^*.$$

The closures of the maximal conic strata having local spans V_1^* and V_2^* are $\mathcal{C}_1^*$ and $\mathcal{C}_2^*$, so they are determined by ν^* . To see why an alternative component cannot mix the two strata, note that a convex cone contained in the union of two incomparable linear subspaces must be contained in one of them. Otherwise it would contain $x \in V_1^* \setminus V_2^*$ and $y \in V_2^* \setminus V_1^*$, while a generic positive combination of x and y lies in neither subspace. Positivity of the fitted component density would then put mass outside $\mathcal{C}_1^* \cup \mathcal{C}_2^*$, contradicting equality of the mixing laws.

There are exactly two fitted components with positive weights, so one must cover each maximal stratum. Neither fitted cone can be a strict subcone, because the corresponding true law has positive density throughout its relative interior. This proves equality of the cones up to a label swap. Restricting the common mixing measure to A_s identifies the subprobability measure $\pi_s^* \nu_s^*$, whose total mass and normalization give π_s^* and ν_s^* separately. Theorem 1 applied to each conditional law then gives the direction-rate and Gamma-splitting conclusion. $\square$

Theorem 5 (Strictly nested cones of different dimensions). *Suppose, after labeling the true components if necessary, that*

$$\mathcal{C}_1^* \subsetneq \mathcal{C}_2^*, \quad V_1^* \subsetneq V_2^*, \quad D_1^* < D_2^*. \quad (15)$$

Then the conclusions of Theorem 4 still hold for any alternative two-component MiSo representation with positive weights and proper, nondegenerate Gamma component laws. The lower-dimensional component is identified as the singular part of the mixing measure relative to D_2^ -dimensional Hausdorff measure on the larger cone.*

Proof. Again the count law identifies ν^* . Since ν_2^* has positive density on $\text{relint}(\mathcal{C}_2^*)$ and $\mathcal{C}_1^* \subset \mathcal{C}_2^*$,

$$\text{supp}(\nu^*) = \mathcal{C}_2^*.$$

Hence the larger cone and its dimension D_2^* are determined by the mixing measure. Let

$$\eta = \mathcal{H}^{D_2^*} \upharpoonright_{\mathcal{C}_2^*}$$

be D_2^* -dimensional Hausdorff measure restricted to that cone. This is a σ -finite reference measure. Lemma 3 gives

$$\nu_2^* \ll \eta.$$

On the other hand,

$$\eta(\mathcal{C}_1^*) = 0$$

because $\mathcal{C}_1^*$ has dimension $D_1^* < D_2^*$, while $\nu_1^*(\mathcal{C}_1^*) = 1$. Therefore $\nu_1^* \perp \eta$.

The Lebesgue decomposition of the finite measure ν^* relative to η is unique [1] and is exactly

$$\underbrace{\nu_{\text{ac}}^*}_{\ll \eta} = \pi_2^* \nu_2^*, \quad \underbrace{\nu_{\text{sing}}^*}_{\perp \eta} = \pi_1^* \nu_1^*. \quad (16)$$

Thus

$$\pi_2^* = \nu_{\text{ac}}^*(\mathbb{R}_+^M), \quad \pi_1^* = \nu_{\text{sing}}^*(\mathbb{R}_+^M),$$

and normalizing the two parts identifies ν_2^* and ν_1^* . Moreover, $\text{supp}(\nu_{\text{sing}}^*) = \mathcal{C}_1^*$, so the smaller cone is identified even though it is hidden in the topological support of the full mixture.

Any alternative proper Gamma-cone representation of the same measure must respect this unique absolutely-continuous/singular decomposition. With two positive components, its higher-dimensional component equals $\nu_{\text{ac}}^*/\pi_2^*$ and its lower-dimensional component equals $\nu_{\text{sing}}^*/\pi_1^*$, up to swapping labels. Applying Theorem 1 inside both identified component laws completes the proof. $\square$

Remark 1 (Anchor words give a direct transform proof). The measure decomposition in Theorem 5 is not needed when there is a nonnegative separating vector. If $v \in \mathbb{R}_+^M$ satisfies

$$\langle v, q_{1d}^* \rangle = 0 \quad \text{for all } d, \quad \langle v, q_{2d}^* \rangle > 0 \quad \text{for at least one } d,$$

then, for every $t \in \mathbb{R}_+^M$,

$$\lim_{u \rightarrow \infty} \mathcal{L}_\mu(t + uv) = \pi_1^* \mathcal{L}_{\mu|Z=1}(t). \quad (17)$$

Taking $t = 0$ identifies π_1^* ; subtraction then identifies the other component transform. An anchor coordinate used by motif 2 but unused by motif 1 supplies $v = e_m$. Geometrically, this includes the important case in which the smaller cone is an exposed face of the larger cone.

4 Why individual overfitted dimensions are not identifiable

The equivalence in Theorem 1 is unavoidable. If

$$\lambda_i^* \sim \text{Gamma}(a^*, b^*)$$

and $a_1, \dots, a_J > 0$ satisfy $\sum_{j=1}^J a_j = a^*$, then independent variables

$$\lambda_{ij} \sim \text{Gamma}(a_j, b^*)$$

satisfy

$$\sum_{j=1}^J \lambda_{ij} \stackrel{\text{d}}{=} \lambda_i^*. \quad (18)$$

Consequently,

$$\lambda_i^* f \stackrel{\text{d}}{=} \sum_{j=1}^J \lambda_{ij} f.$$

A likelihood-based procedure cannot distinguish one Gamma dimension from several dimensions sharing its direction and rate. Only the total shape is identified. This common-rate additivity is the elementary finite-dimensional form of the infinite divisibility of the Gamma law [8].

Remark 2 (Merging and truncation are different). Theorem 1 assumes proper Gamma distributions with finite positive shape and rate. Under this parameterization, a surplus dimension cannot be exactly null. It must duplicate a true direction-rate pair. Exact truncation requires enlarging the parameter space to include a boundary such as $a_h = 0$, $b_h = \infty$, or an explicit activity indicator. Empirical-Bayes fitting may approach such a boundary, but it may instead split a true shape as in (18).

Remark 3 (Geometric and statistical dimensions). If the f_d^* are linearly independent, the cone

$$\mathcal{C}_* = \text{cone}(f_1^*, \dots, f_{D_*}^*)$$

is simplicial and has geometric dimension D_* . Theorem 1 actually needs only distinct scaled directions. Hence the Gamma distribution can distinguish a positive-activity direction even when that direction lies inside the cone generated by other directions: it contributes its own singular hyperplane to the log-Laplace transform.

This distinction is important relative to standard NMF theory. Generic NMF need not be unique, and its identifiability is normally formulated through the geometry of all feasible nonnegative factorizations [6, 2]. Here the normalized cone geometry removes the usual scaling ambiguity, while the Gamma loading law supplies information beyond the first-moment matrix. The resulting statistical identification can therefore hold under conditions different from deterministic NMF uniqueness.

5 Factorial-cumulant interpretation

The same structure appears in the joint factorial cumulants of the observed counts. For order $r \geq 1$, let $\mathcal{K}_r^{(F)}(Y)$ denote the order- r factorial-cumulant tensor. Conditional Poisson sampling implies that it equals the ordinary cumulant tensor of the random mean. Gamma cumulants therefore give

$$\mathcal{K}_r^{(F)}(Y) = (r-1)! \sum_{d=1}^{D_*} \frac{a_d^*}{(b_d^*)^r} (f_d^*)^{\otimes r}. \quad (19)$$

Thus the full sequence of factorial cumulants contains the same direction, rate, and shape information as the Laplace transform. Equation (19) may also provide a constructive route to initialization or diagnostics, although the transform argument gives the cleaner identifiability proof. The equality between factorial cumulants of a mixed Poisson variable and cumulants of its mixing variable is standard [4, Sec. 8.3.2]. Recovering the rank-one terms from one or several such tensors connects this constructive route to classical tensor-decomposition uniqueness theory [5].

6 Implications for MiSo post-processing

Within every identified subcone, the population target should be defined as the set of distinct Gamma direction-rate pairs, not as the raw number of fitted dimensions. Theorems 4 and 5 first align the mixture components, after which Theorem 1 suggests:

1. identify dimensions with negligible fitted loading activity and truncate those approaching the null boundary;
2. group active dimensions with nearly equal normalized factor directions and nearly equal Gamma rates;

3. merge every such group by summing its Gamma shapes; and
4. report the number of retained groups as $\hat{D}_*$.

In the full MiSo model, the generative direction of dimension d is a selected factor row F_{γ_d} . The variational probabilities $\bar{\gamma}_{dk}$ quantify uncertainty about that selected direction. They can reveal competition among duplicate or nearly duplicate candidate factor rows, while Theorem 1 explains why an active, genuinely different direction cannot persist in an exact overfitted population representation. The theorem concerns the generative model; finite-sample variational fits may still display local optima or approximate splitting and should be assessed by predictive fit and stability across initializations.

7 Next theoretical step

The single-subcone result separates two questions that are confounded in a raw overfitted fit:

null dimensions versus Gamma splits of active dimensions.

For $S = 2$, the remaining difficult case has equal-dimensional component laws living in the same span with overlapping relative interiors. Neither conic stratification nor Hausdorff dimension then separates the two measures. One must instead prove uniqueness of the decomposition of the transform as a sum of two Gamma-product transforms. Classical finite-mixture identifiability results [10, 11] provide a natural starting point. A later extension to $S > 2$ must additionally align multiple conic strata and handle repeated or overlapping component laws.

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